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eva3dm: A R-package for model evaluation of 3D weather and air quality models

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Software

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Summary

Eva3dm is a package designed to support the evaluation of 3-dimensional physical models (particularly, weather and air quality models) against observation data in order to quantify different errors and bias present in the model results.

Statement of need

Evaluation is the most important step for any model application, it can assure that the model results represent accurately the interest variables. There are currently other tools available in R (Carslaw & Ropkins, 2012), Python (Ladwig, 2017) or other languages (Appel, 2011; NCAR et al., 2017) but none provide the degree of integrated support presented in this package: from the pre-processing of observations and model output, to evaluation and visualization.

Description

The literature presents various evaluation criteria depending on the variable (Emery, 2001, 2017; Monk, 2019; Ramboll, 2018; Zhai, 2024; Zhang, 2019), which can be used to compare models and assess their performance. These criteria vary based on the simulation goal, observation variability, and measurement errors.

A brief description of the steps to perform a model evaluation is provided below:

1. **Pre-processing of observations**:
   - Download of the observations, some examples include:
     - METAR can be downloaded using the R-package riem or the Iowa State University site
     - AERONET can be downloaded at AErosol RObotic NETwork site
     - Air Quality data for Brazil can be downloaded using the R-package qualR, or QUALAR and MonitorAir sites
     - Satellite products are available at NASA giovanni website
   - Process a list of the location to extract time-series from the model.
   - Process observation data for evaluation: Unit conversion, time zone conversion to UTC, and calculation of secondary variables.
   - Quality Assurance of the observation data: check for values outside the valid range, check if the data is available for the time-period and region of the simulation and note any singular event.

2. **Pre-processing of model output**: Extraction of model outputs, unit conversion and calculation of secondary variables.

3. **Model Evaluation**: The evaluation involves pairing observations with model results and calculating the statistical and/or categorical indices.

4. **Visualization**: Visualization of model results and statistical results.

To support these steps, following functions are available:

Pre-processing of observations:

The functions `rh2q` and `q2rh` convert humidity units and the functions `mda8`, `ma8h`, `hourly`, and `daily` can be used to calculate average of time-series. The format used to evaluate time-series is a data.frame, the first column must contain time (in POSIXlt) and one additional column for each different location, satellite data can be read using the function `rast` from R-package terra.

Pre-processing of model outputs:

The function `extract_serie` extract and save time-series from model outputs using a data.frame with name (row names) and latitude (column lat) and longitude (column lon), while the functions `extract_mean` and `extract_max_8h` extract the average or the daily maximum of 8-hour moving average and save in a new NetCDF file;

The function `wrf_rast` can be used to read model output and return a `SpatRaster` or `SpatVector` object from the model files and its counterpart `rast_to_netcdf` that converts a `SpatRaster` to an array and/or save to a existing NetCDF file;

The functions `uv2ws` and `uv2wd` can be used to calculate wind speed and velocity from the model wind components (u and v) and the function `rain` can be used to calculate hourly precipitation from model accumulated precipitation variables.

Model evaluation functions:

There are two high level evaluation functions implemented in the package: `eva`, that does the temporal pairing of both model and observations by station (or combine all data) and the data of time-series and `sat` that interpolate and pair data in regular grids, these functions call the low level evaluation functions `stat` to calculate the statistical metrics or the `cate` to calculate categorical metrics in relation to a threshold value. These result can be written and read using the `write_stat` and `read_stat` functions.

Visualization and extractting information functions:

There are functions for visualization, interpolation and to extract information from NetCDF files, Table 1 show a relation of the visualization functions.

Table 1: Visualization, interpolation and information functions.

Function name	Description
`plot_rast`	Custom plot for `SpatRaster` objects
`plot_diff`	Custom plot for absolute or relative difference of two `SpatRaster` objects
`overlay`	Custom plot to overlay points
`legend_range`	Custom legend that displays max, min and average
`interp`	Interpolation function that combines project and resample for `SpatRaster` objects
`ncdump`	Print a `ncdump -h` equivalent command for a NetCDF file

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Function name	Description
`vars`	Return the name of the variables on NetCDF file
`atr`	Read and write attributes from a Netcdf file

Figure 1 shows examples of the first 4 functions on Table 1:

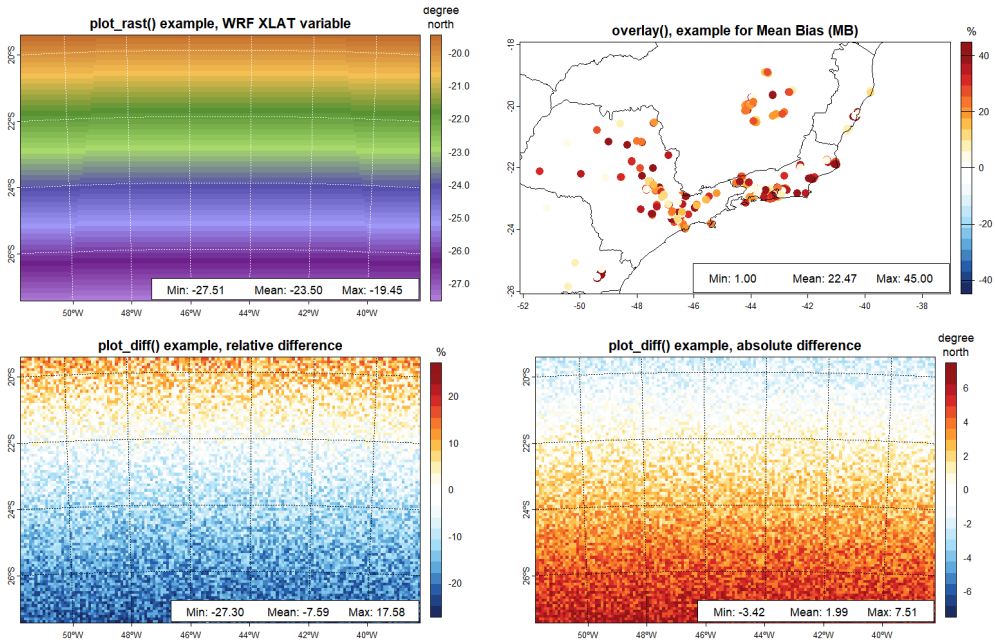

Figure 1: Figure 1 - Example from the viasualization functions.

Speciall functions:

Table 2: Special functions.

Function name	Description	Objective
`%at%`	Combine a data.frame containing evaluation results and a data.frame containing geographical coordinates (site list)	To georeference and visualize the statistical results
`%IN%`	Filter a observation data.frame based on model time-series data.frame. Also can be used to crop a `SpatRaster` based on a second `SpatRaster`	To perform fair comparison of different simulations
`template`	Create folders, post-processing and evaluation scripts	To allow quickly process and evaluate multiple variables from one or multiple simulations

Note that the examples from `eva3dm` are focused on the Weather Research and Forecasting coupled with Chemistry WRF-Chem (Grell, 2005), but but the package can be applied to other models changing some of the arguments from the package functions.

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